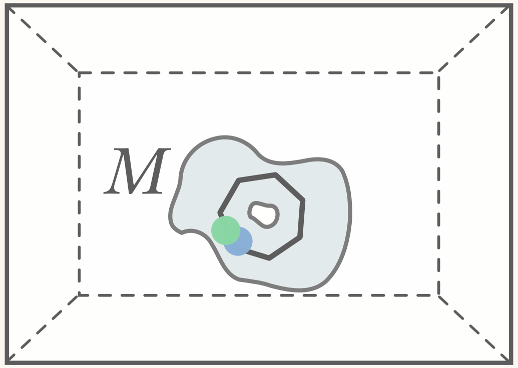
Noncommutative Topology in Quantum Error Correction



David Wakeham

`david@torsor.io`

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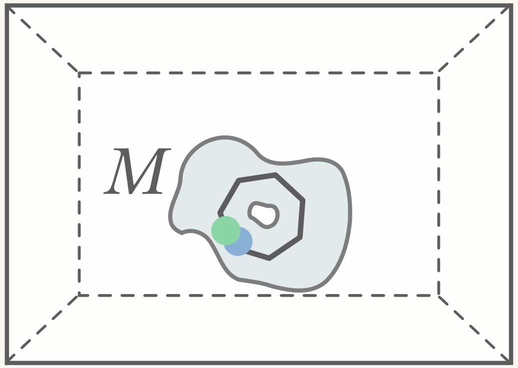

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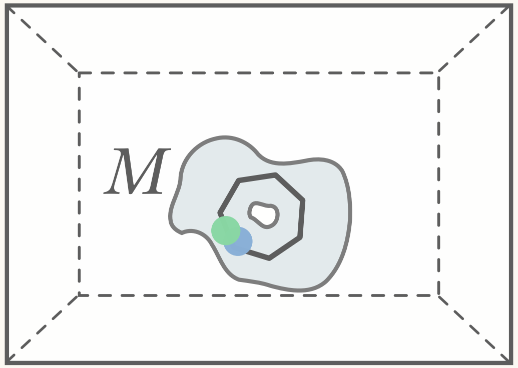

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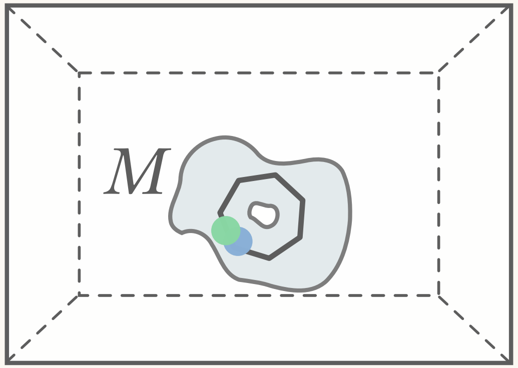

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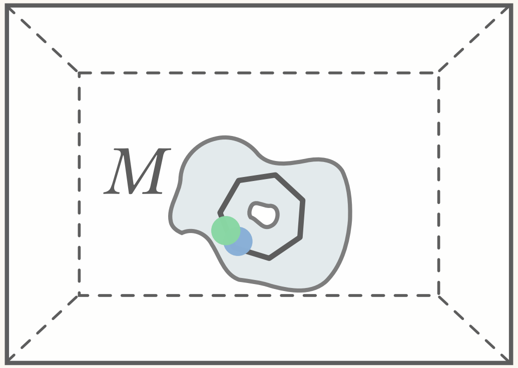

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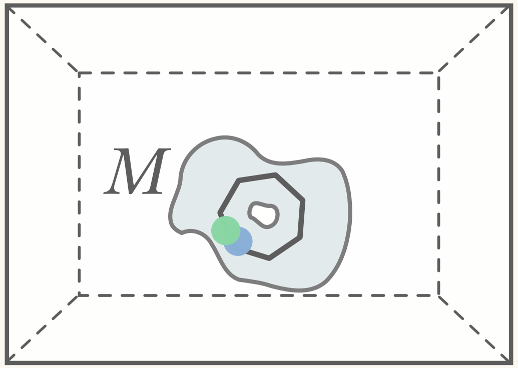

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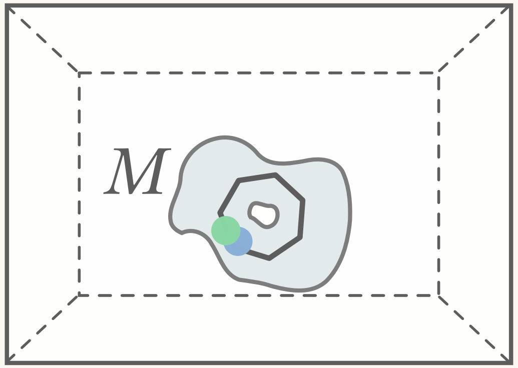

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  - explain why topology is relevant;[GZ = Gottesman-Zhang 2013]
  - explain why noncommutative geometry is relevant;[KS = Kanno-Shimada 2026]
  - combine them and try to compute stuff.[DSBW = Doll, Schulz-Baldes, Waterstraat 2023; CL = Cerjan-Loring 2024]

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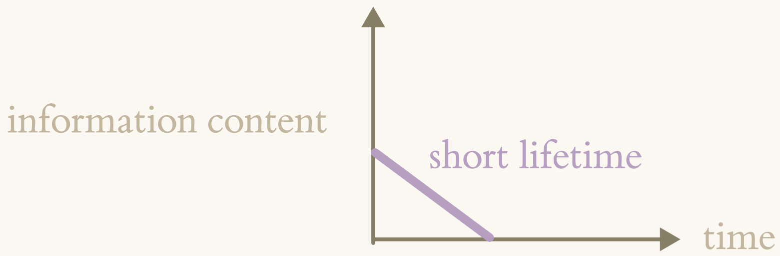

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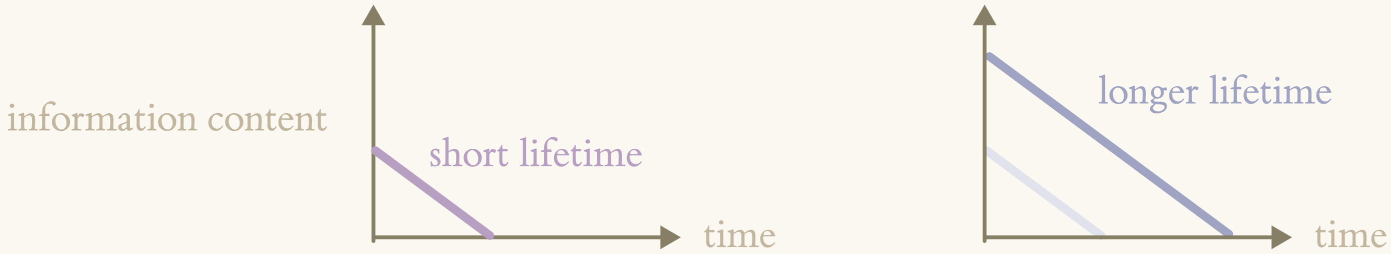

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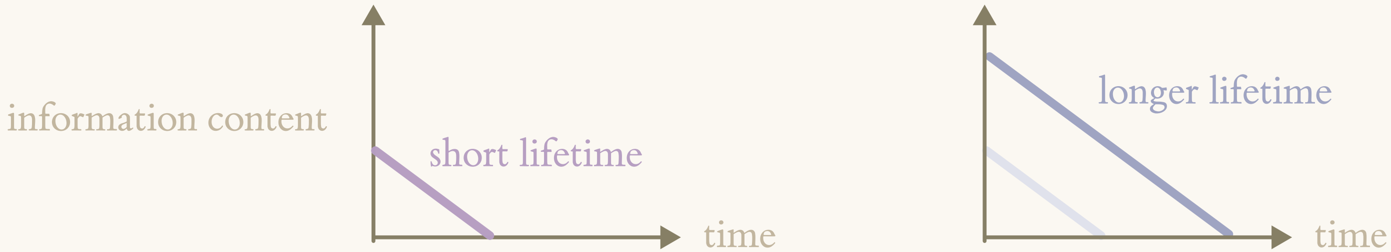

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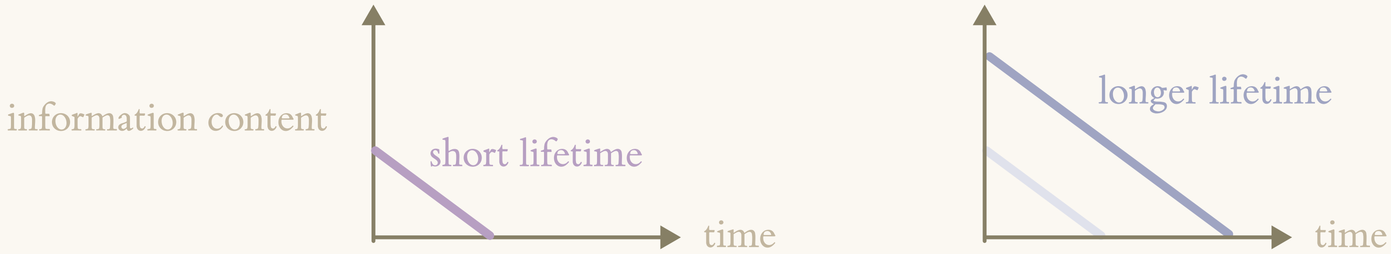

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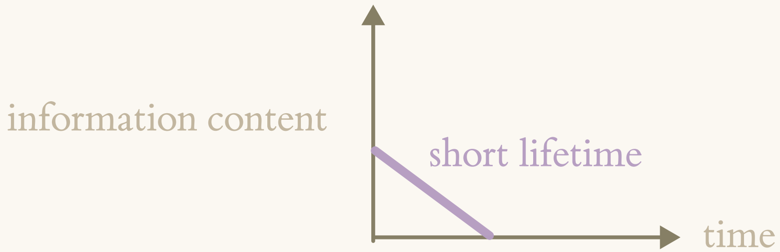

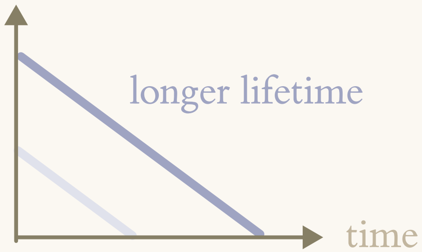

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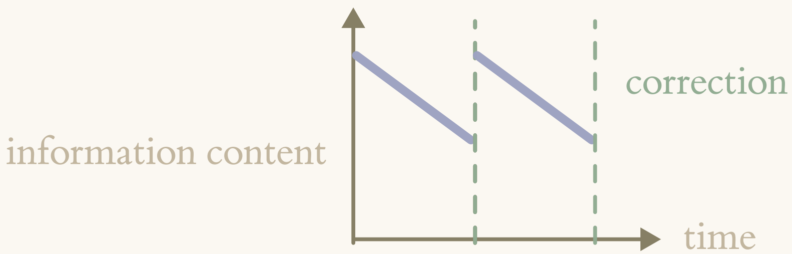

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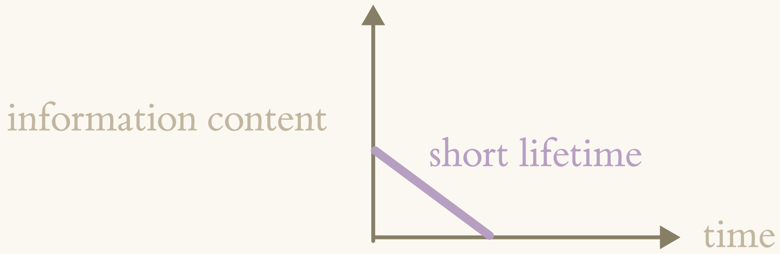

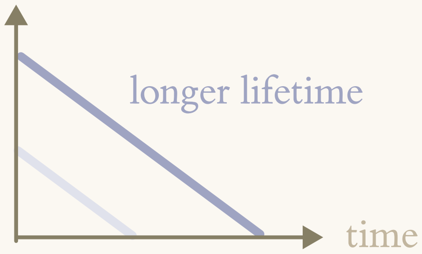

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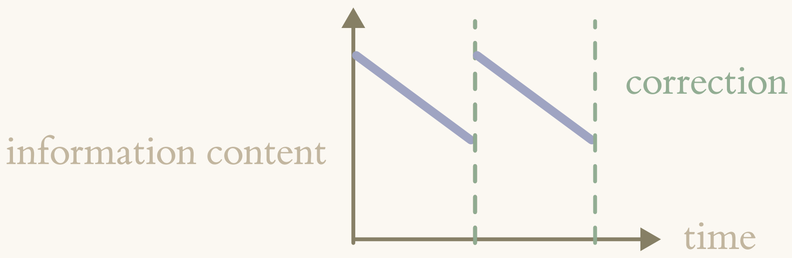

- Analogy to materials: we're trying to engineer collective quantum behaviour.

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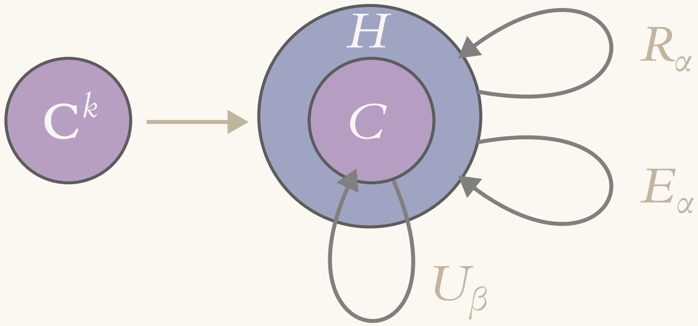

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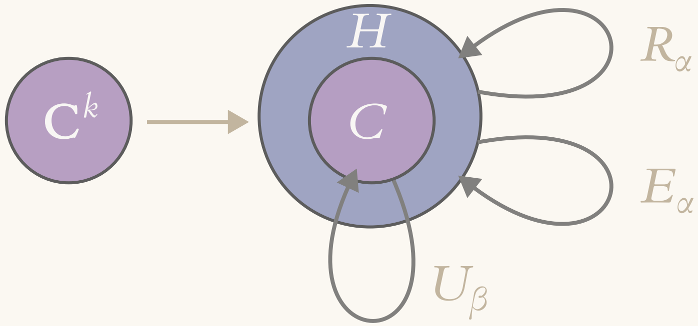

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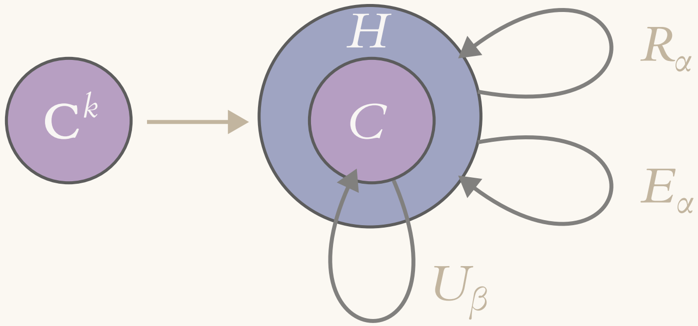

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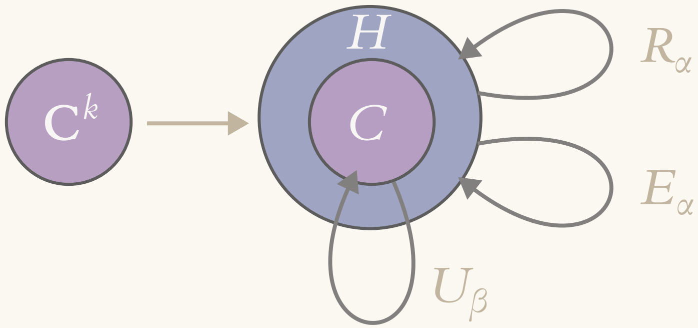

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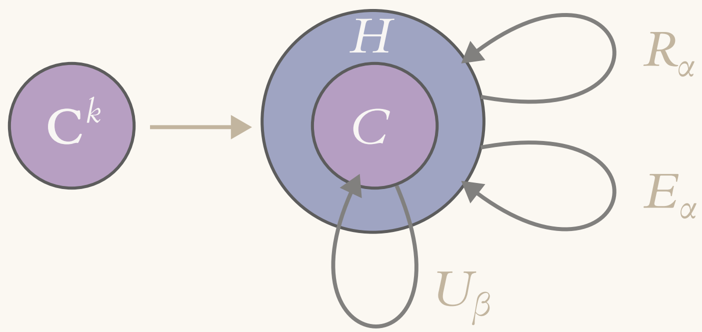

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  - the U_β can be viewed as acting on H, or embedded unitaries acting on $\mathbf{C}^k$.

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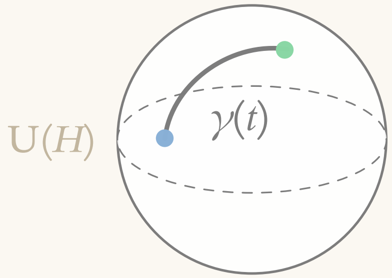

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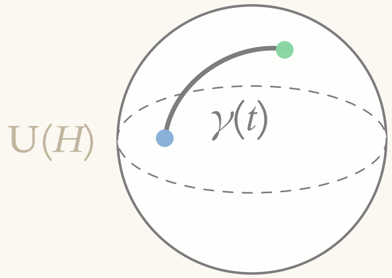

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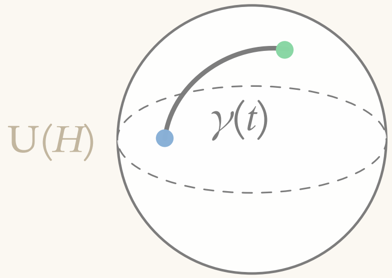

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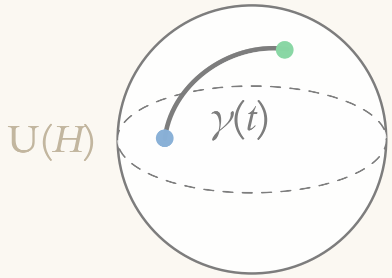

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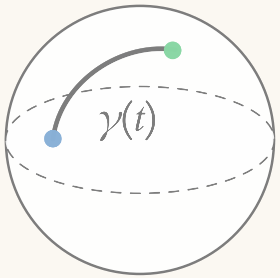

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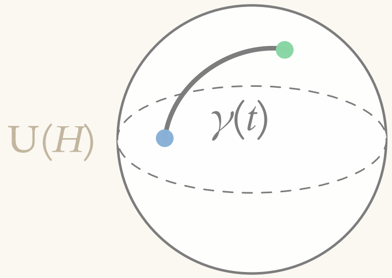

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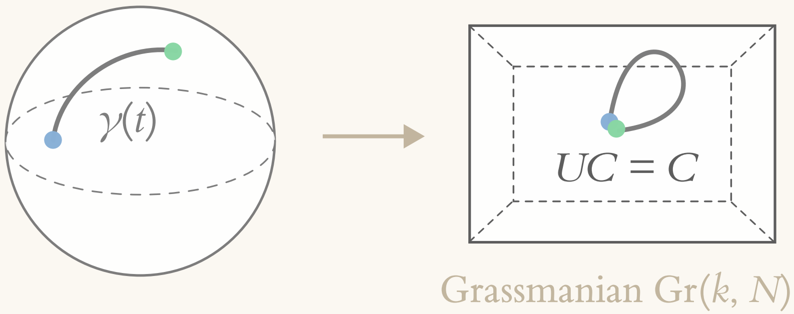

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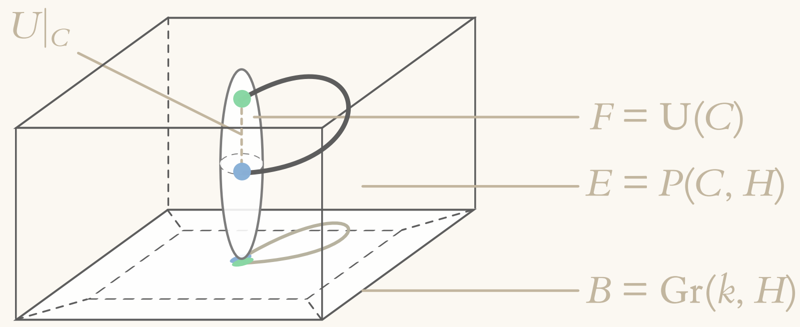

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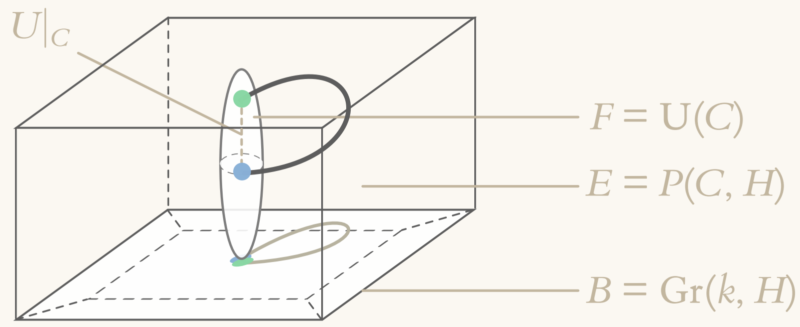

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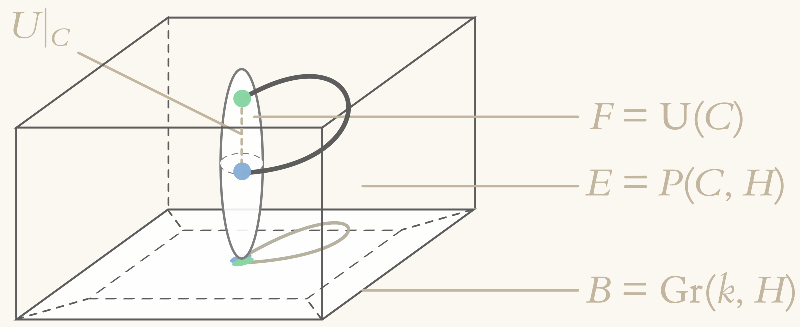

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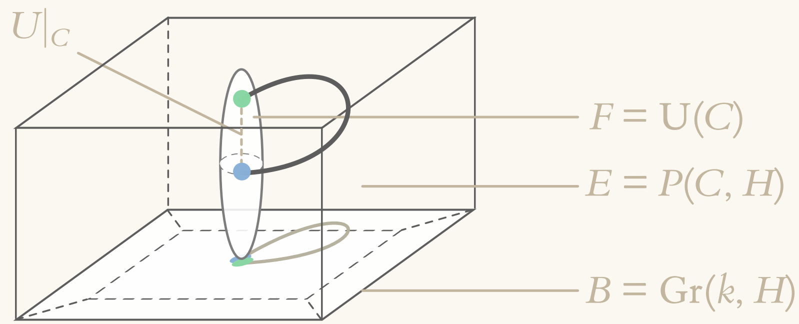

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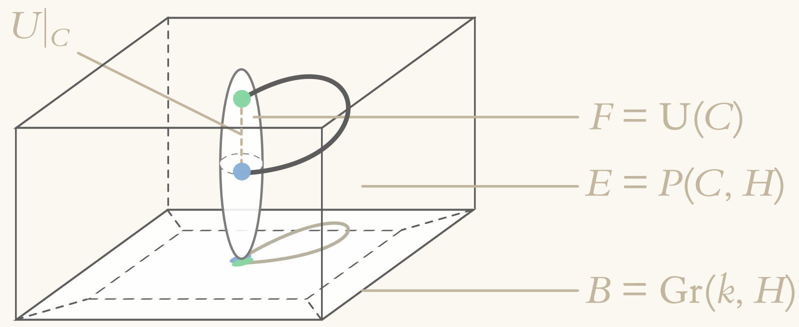

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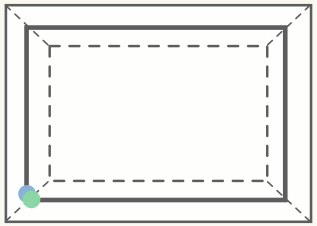

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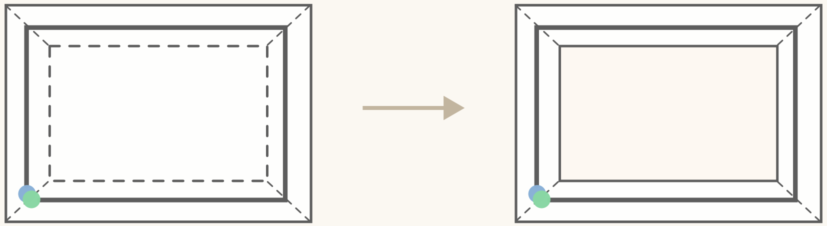

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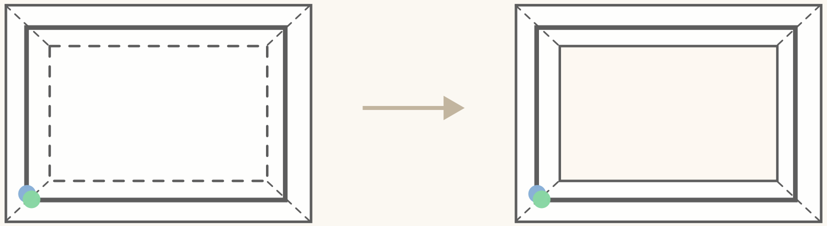

  - the restricted bundle $P(C, H)|_M$ is *projectively flat*, $\gamma_{\mathrm{null}}(1)C = \lambda C$; and

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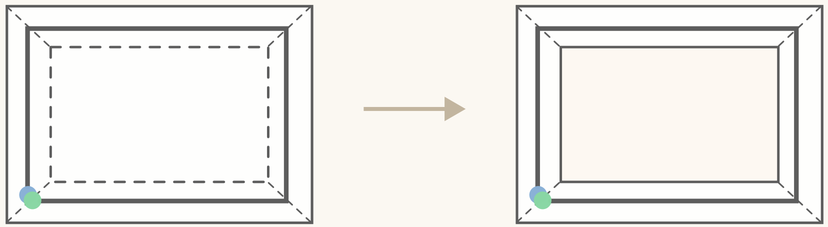

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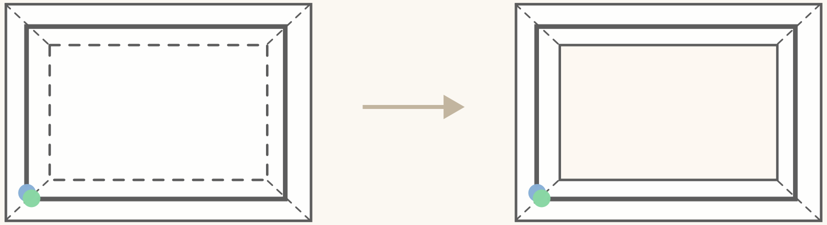

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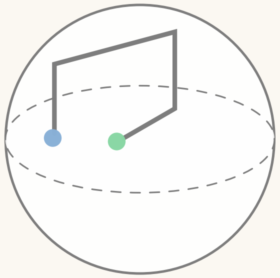

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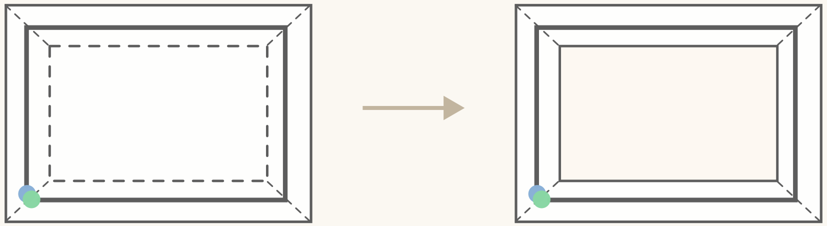

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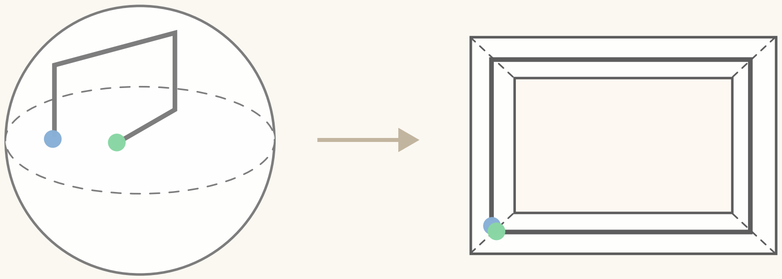

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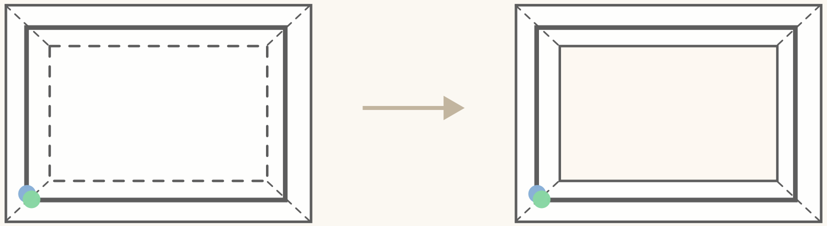

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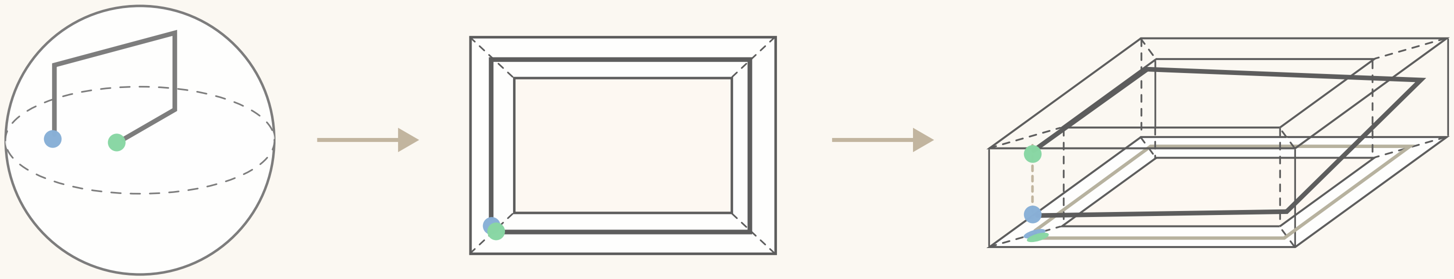

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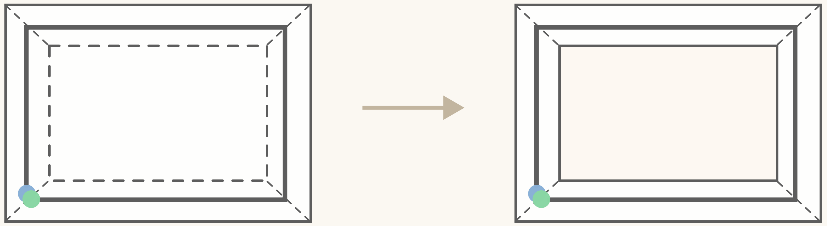

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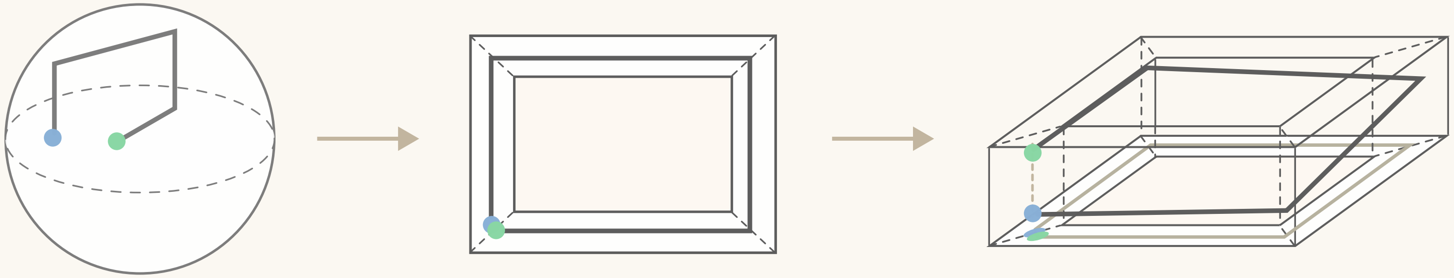

- It's a nice story, and it works for stabilizers, the toric code, and GKP!

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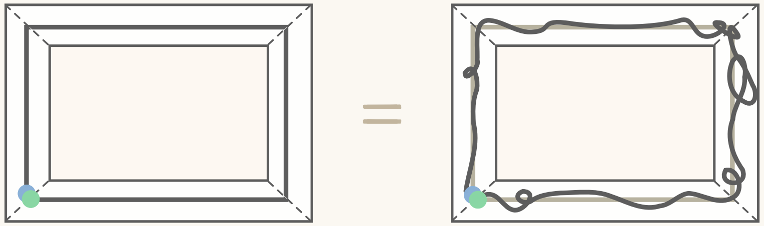

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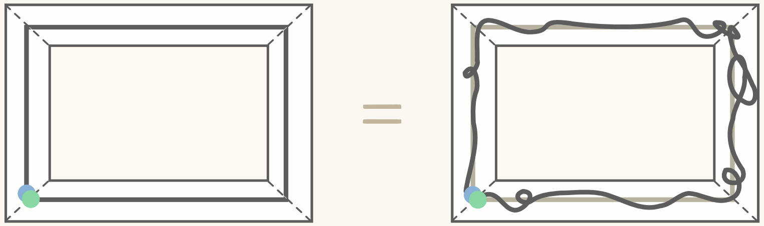

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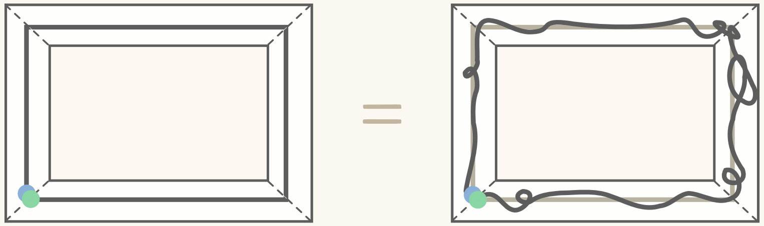

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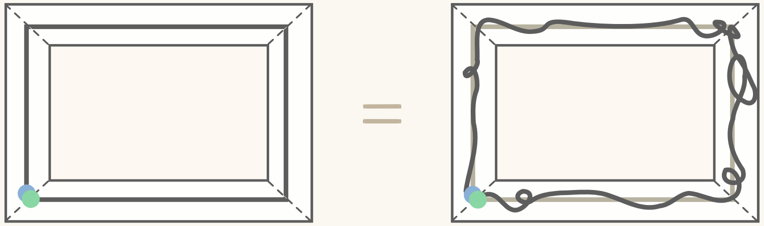

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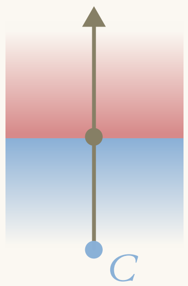

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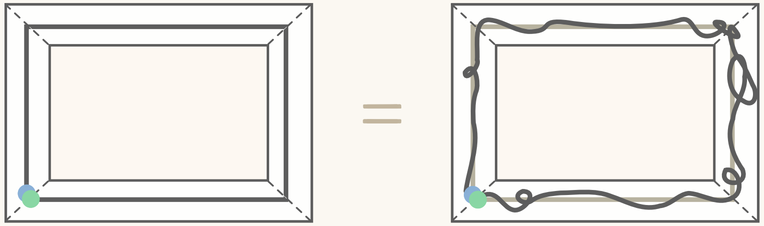

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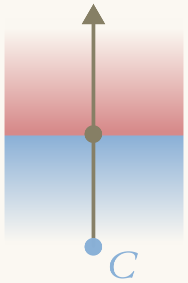

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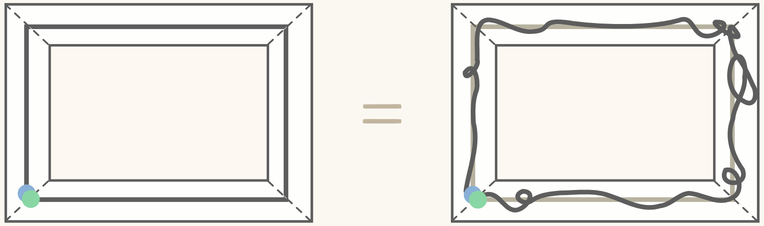

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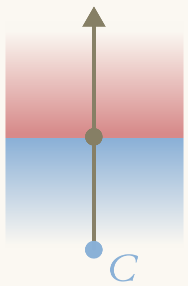
code subspace : Hamiltonian 

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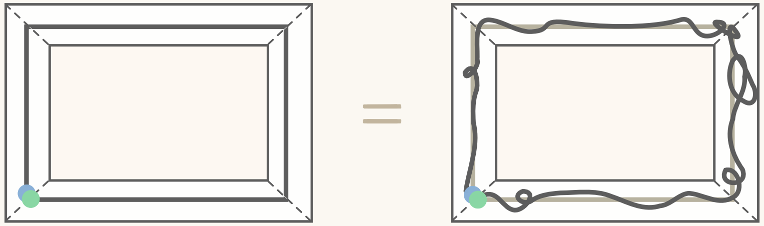

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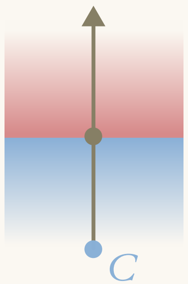
code subspace : Hamiltonian 

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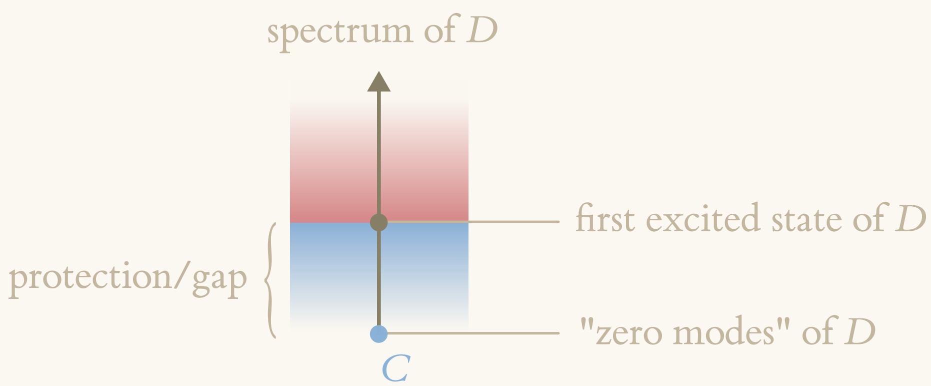

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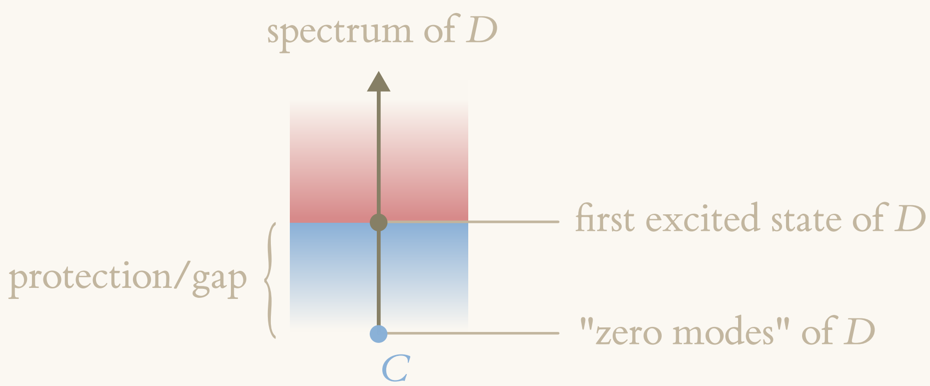

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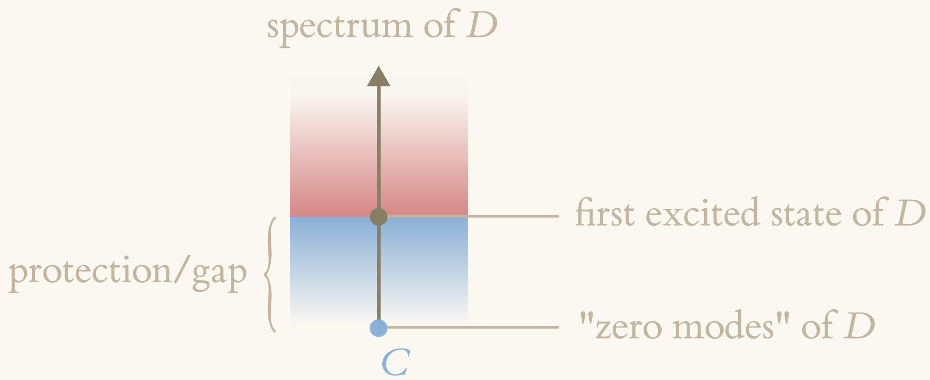

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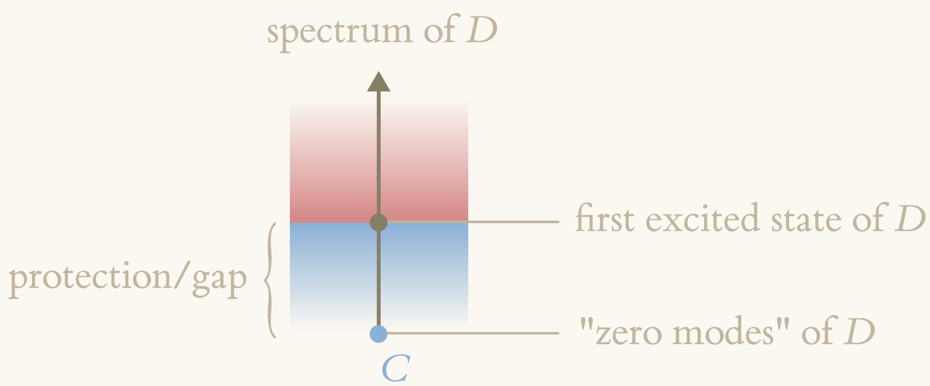

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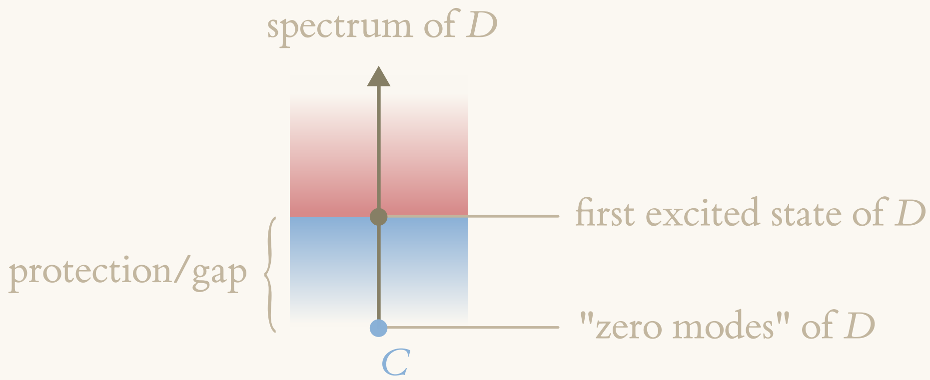

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- We interpret D as a Dirac operator for the triple and note it is gapped!

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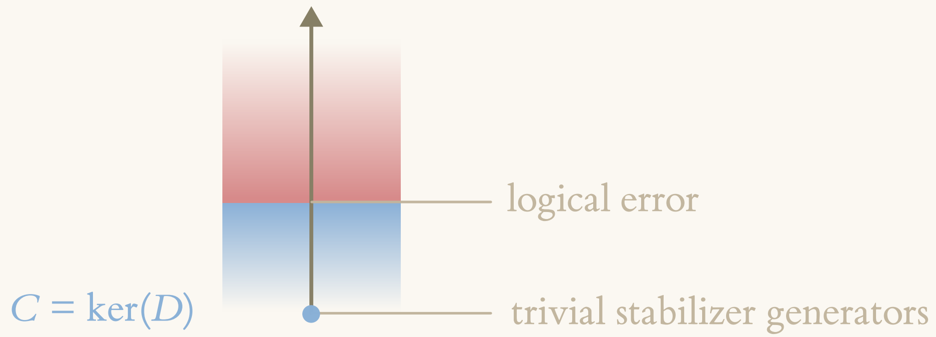

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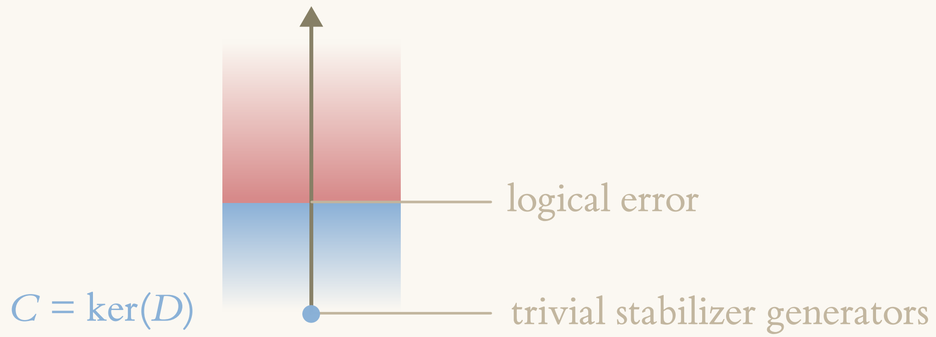

- Note: we have a projective unitary representation $U_\sigma(g)$ with phase (cocycle) σ, and which is trivial on the stabilizer. These are conditions that will reappear below!

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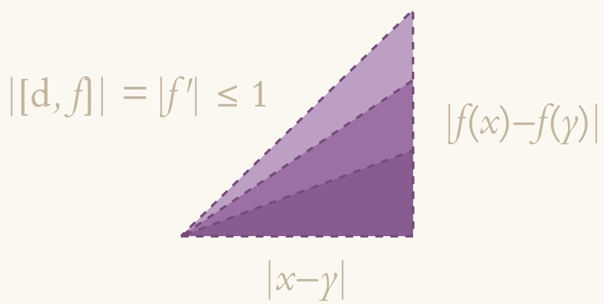

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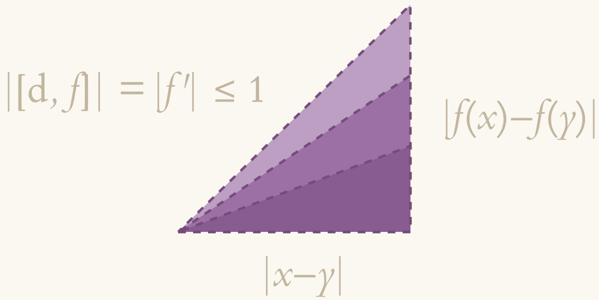

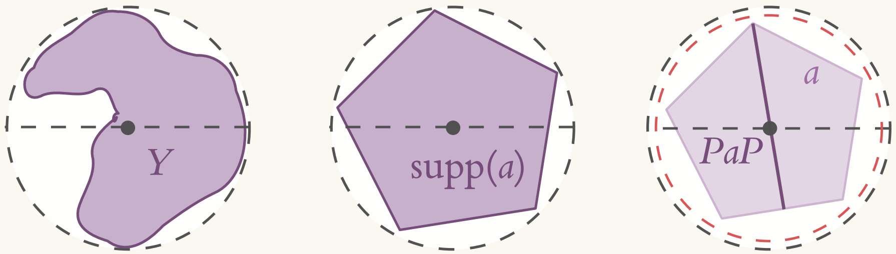

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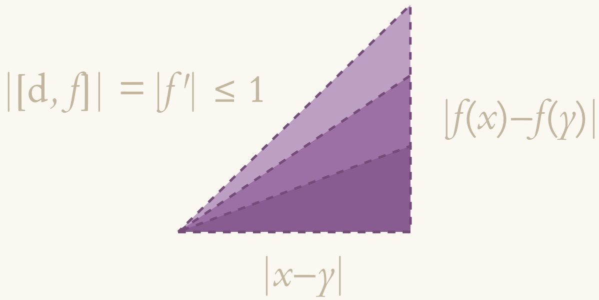

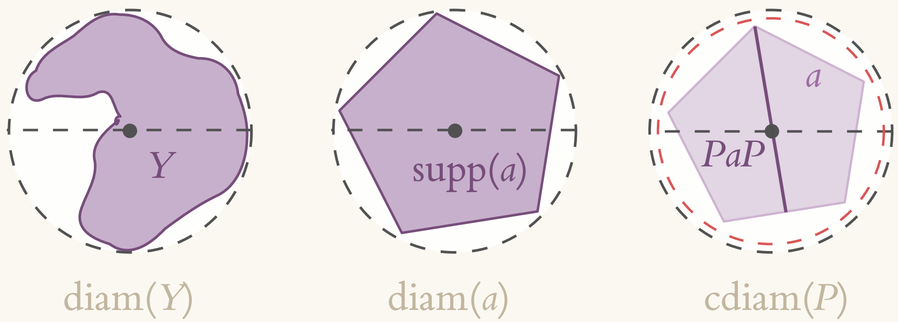

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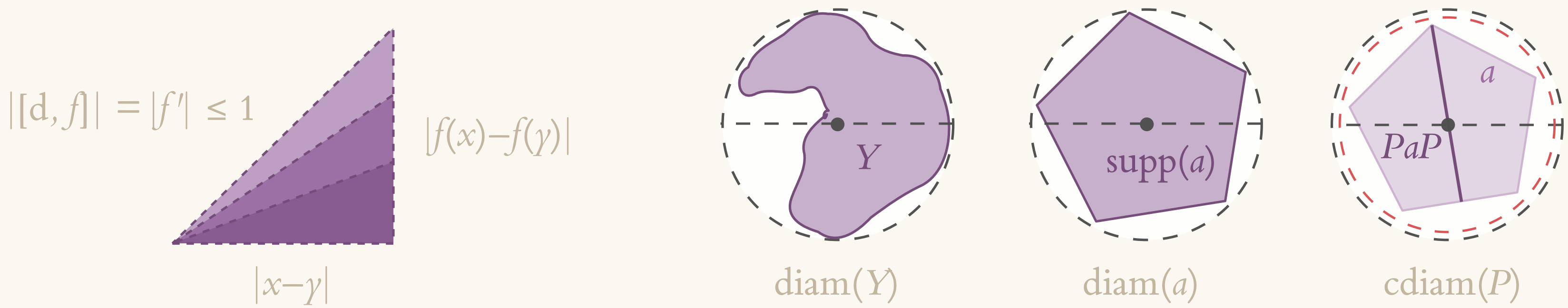

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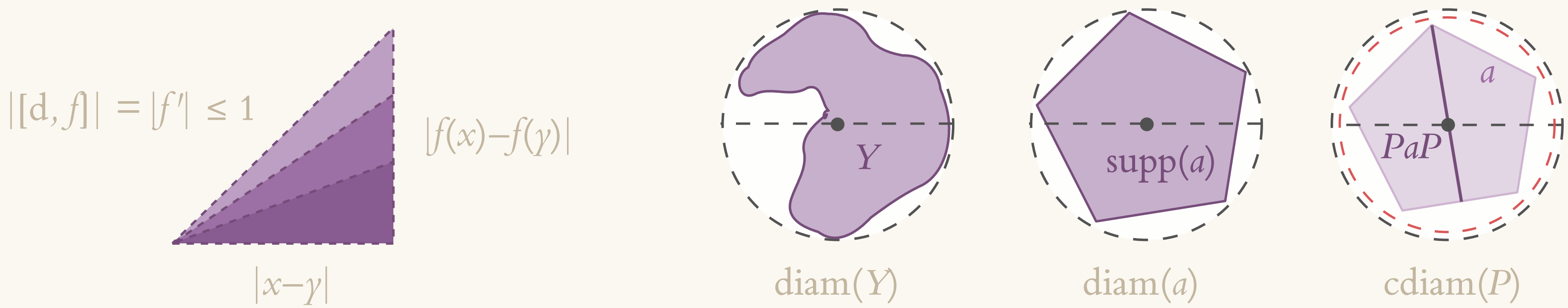

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By a famous theorem of Knill-Laflamme, such errors are correctable.

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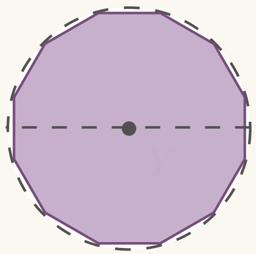

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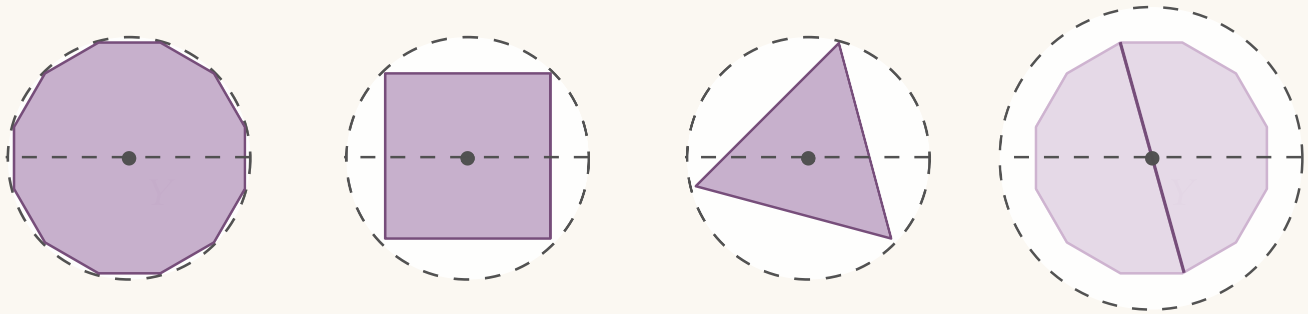

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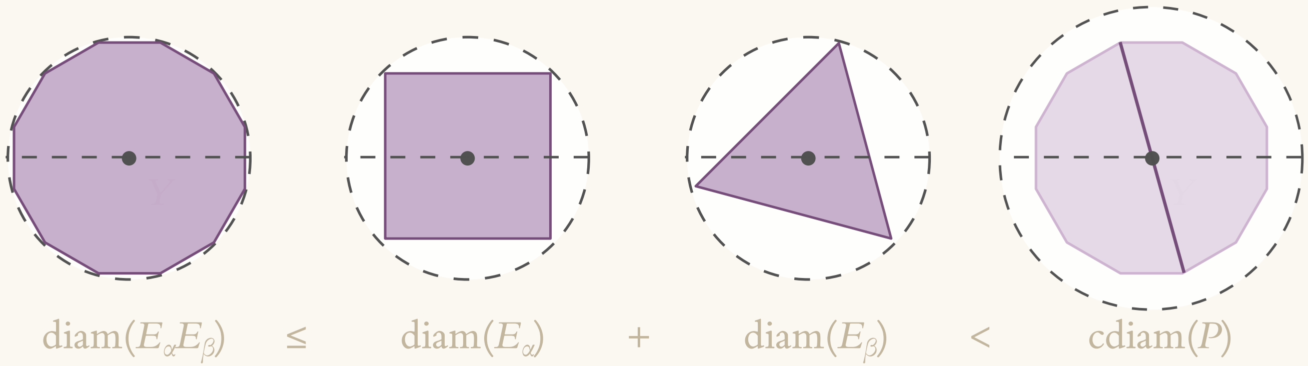

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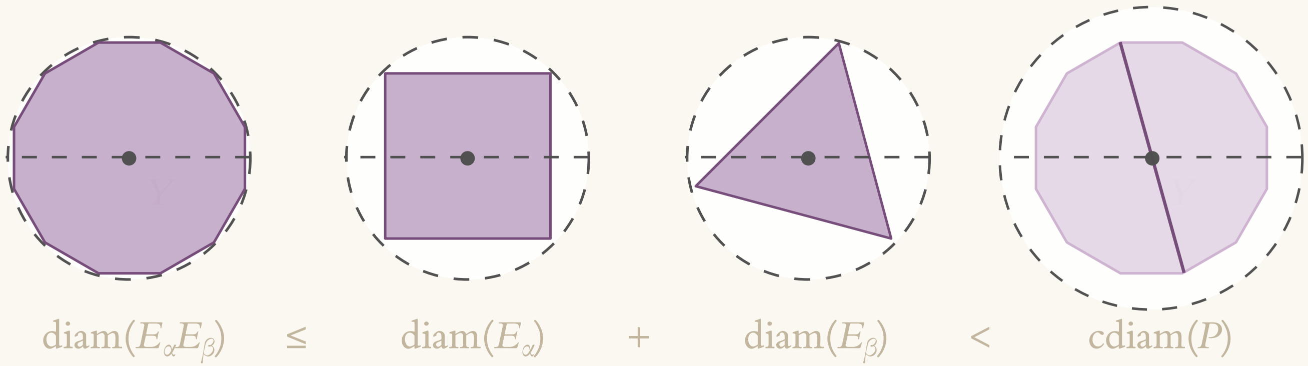

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- So error correction is closely tied to noncommutative geometry! Hopefully we can tie this back to the bundle picture and deformation-resistant operations.

Spectral bundles I

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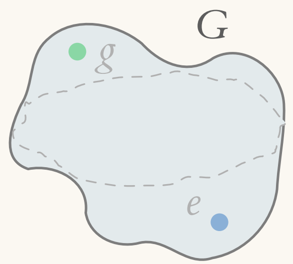

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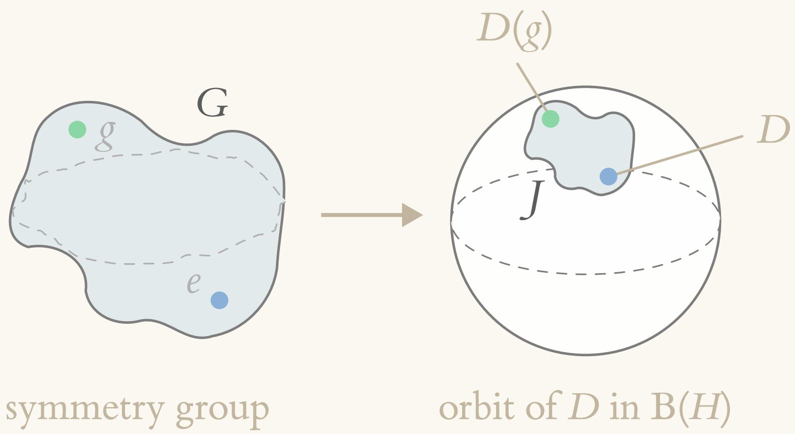

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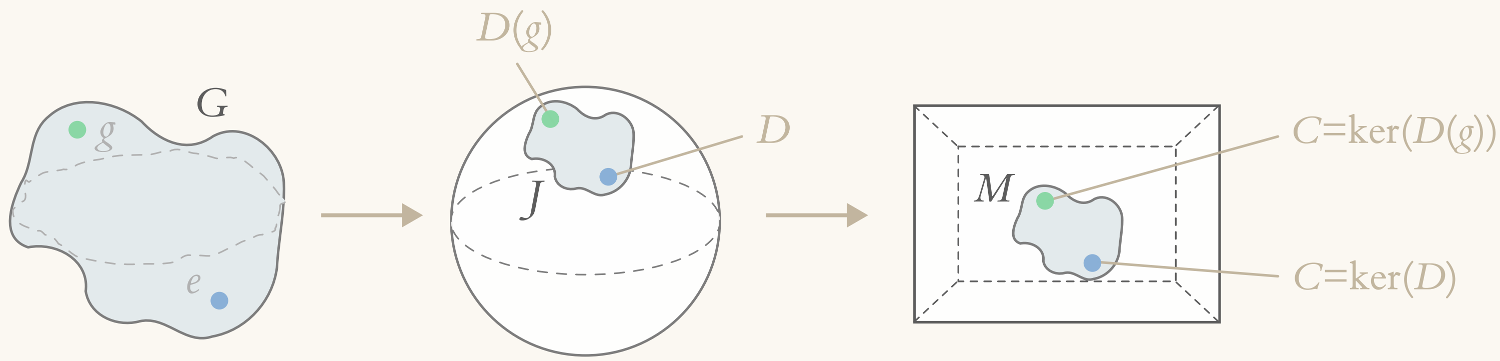

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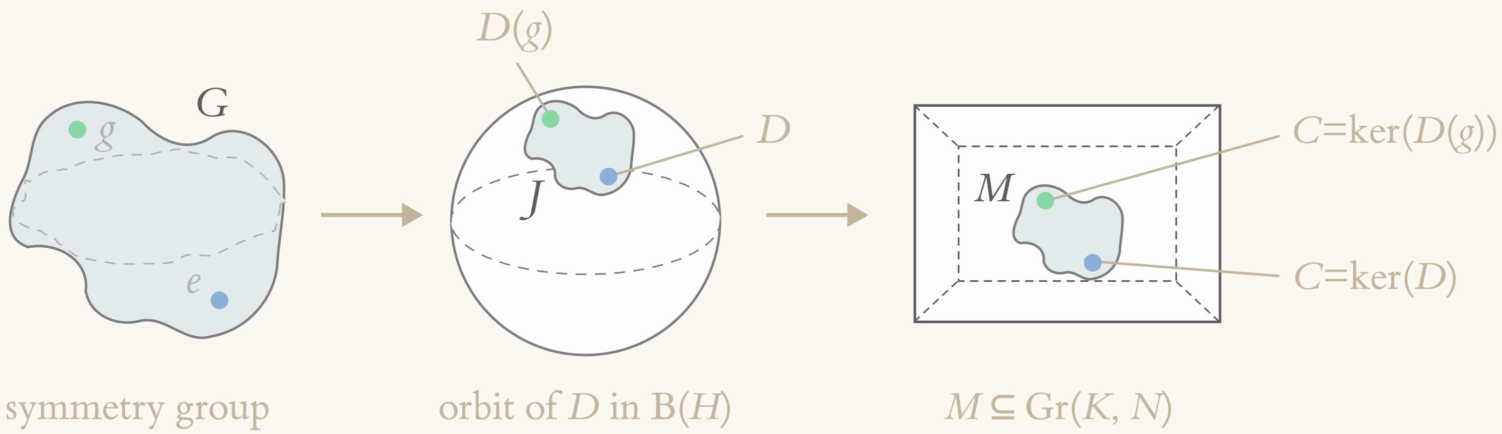

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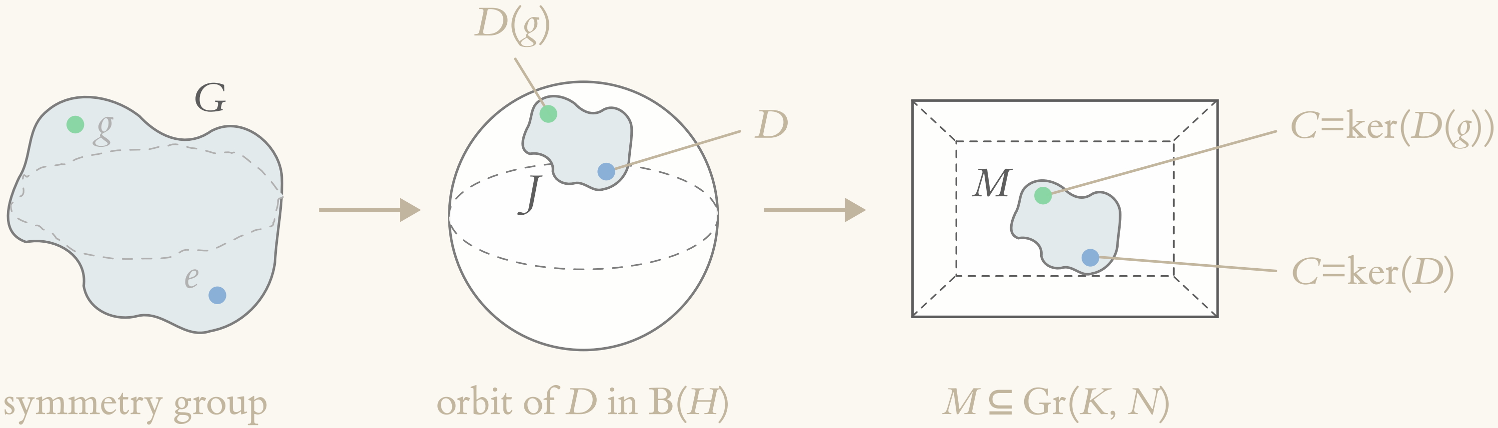

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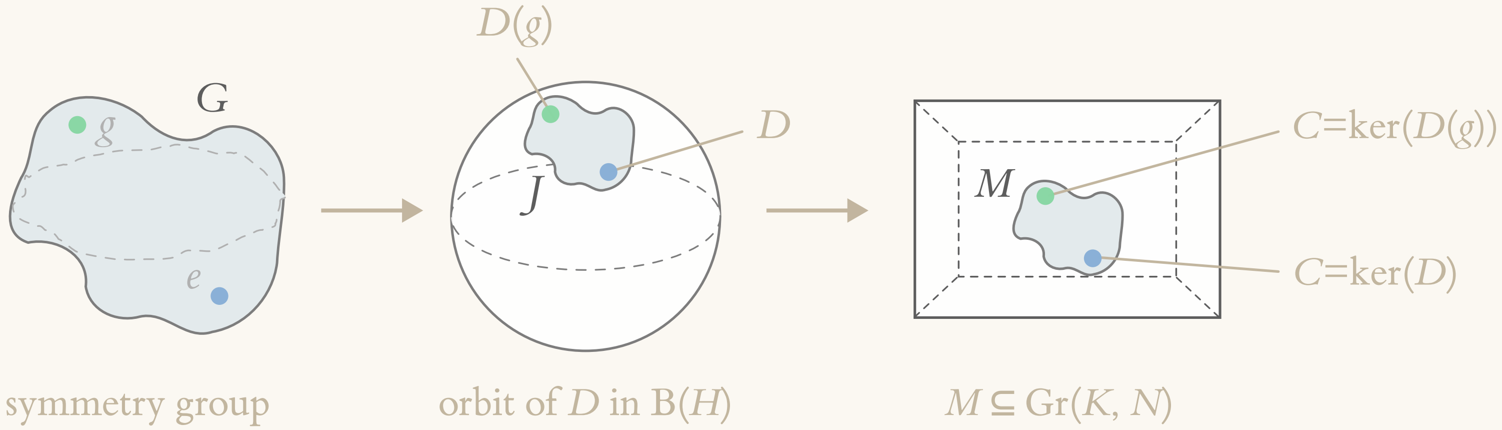

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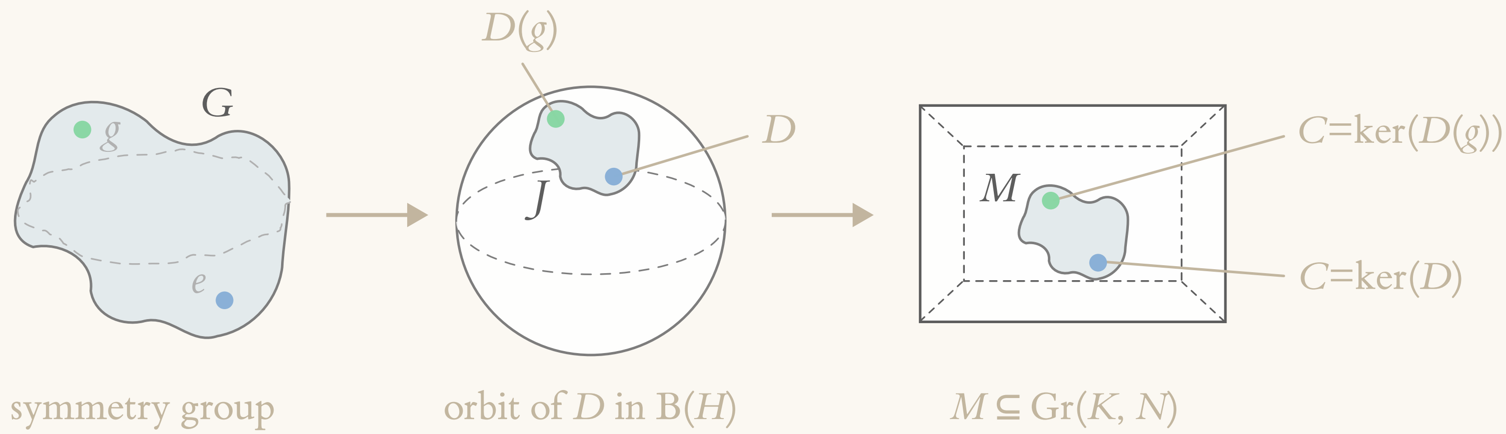

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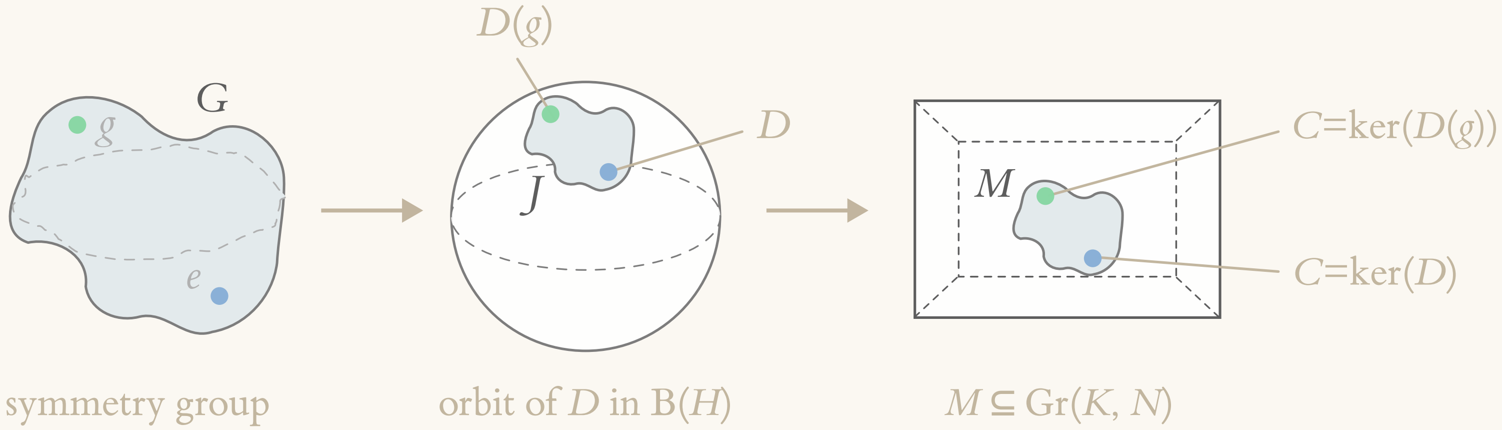

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  - the union of kernels is a smooth submanifold $M = \ker(G \cdot D) \cong G/S_D$.

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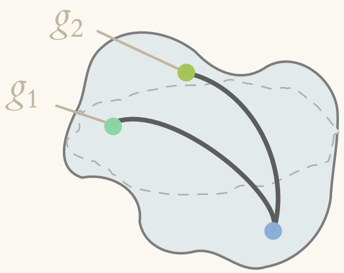

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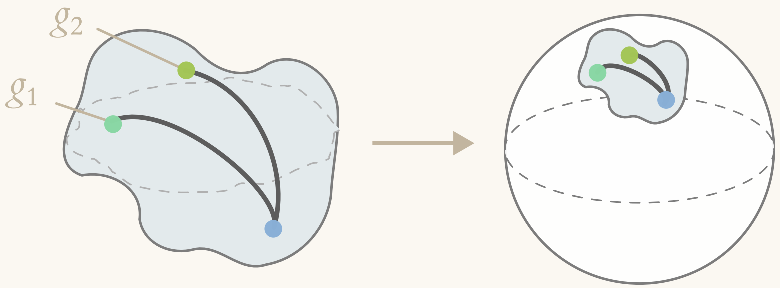

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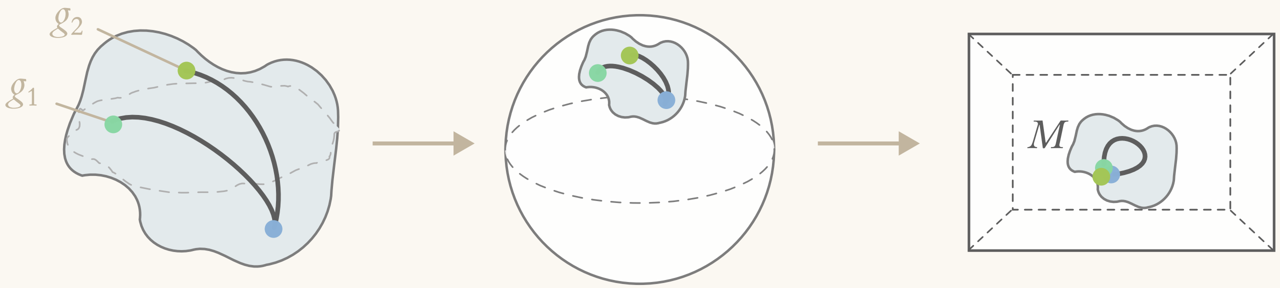

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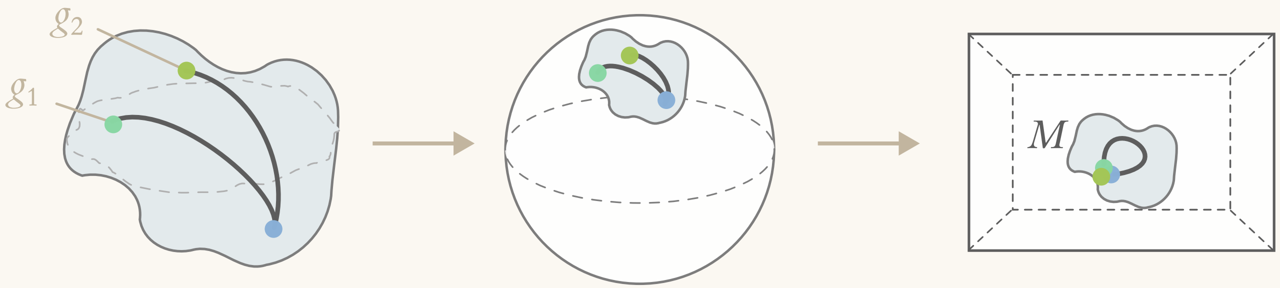

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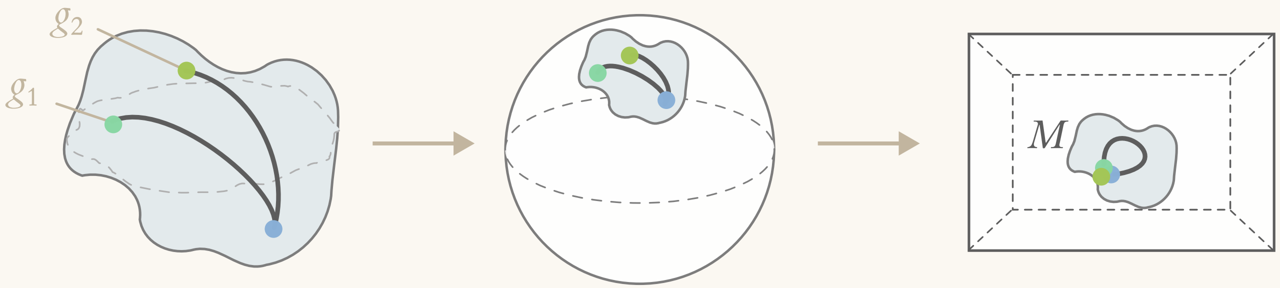

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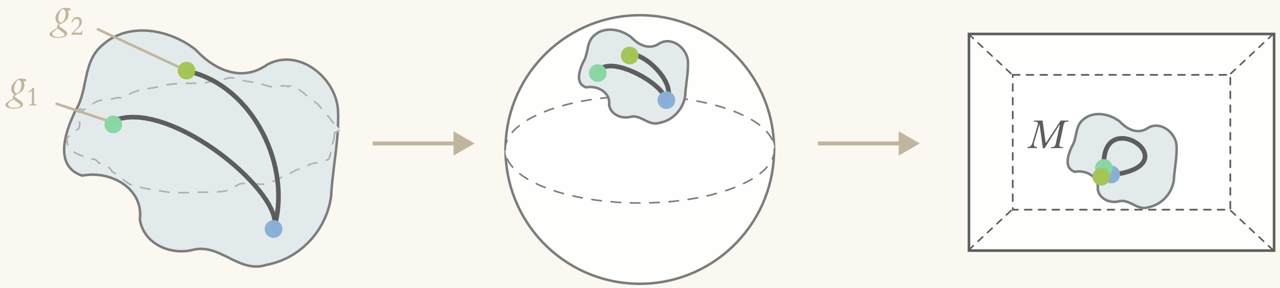

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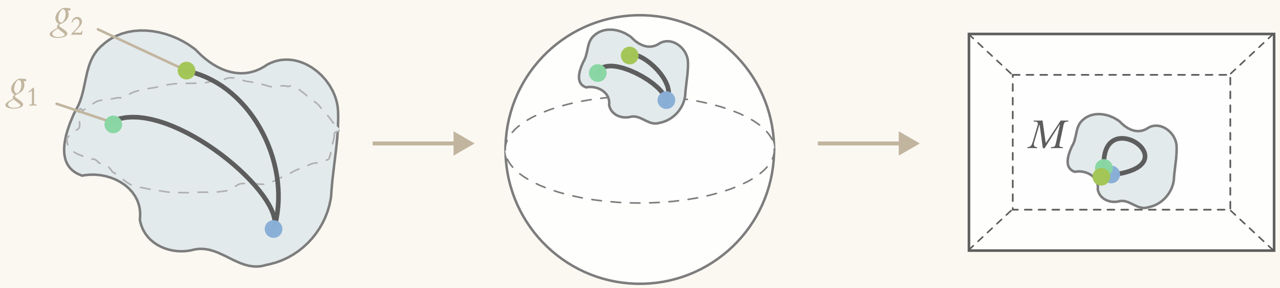

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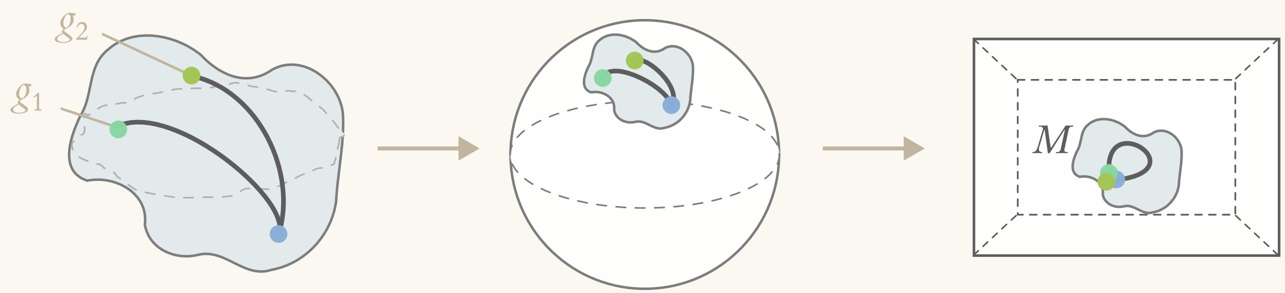

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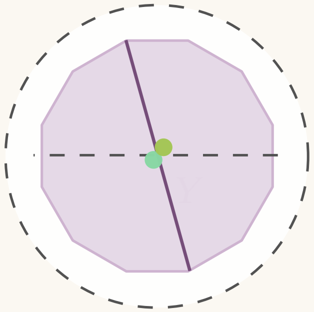

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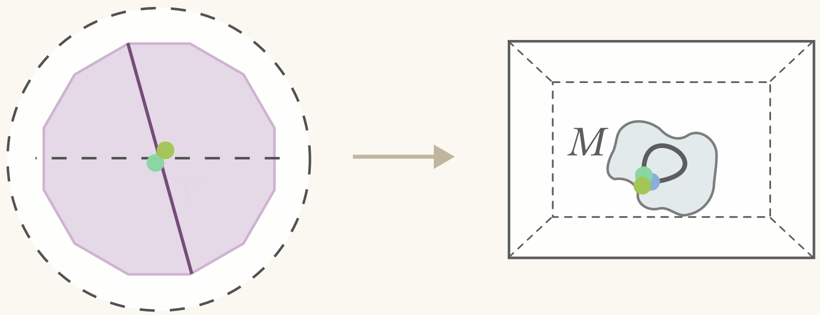

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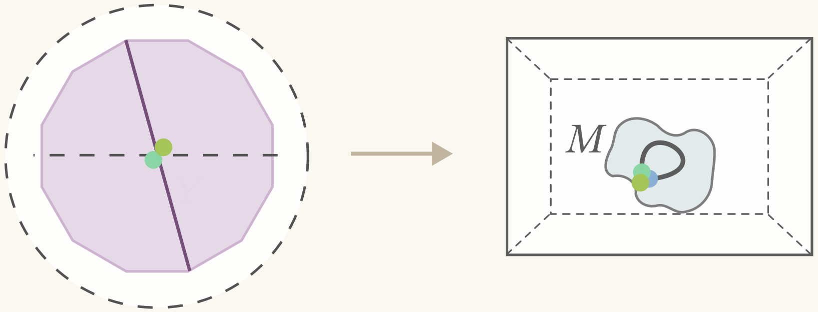

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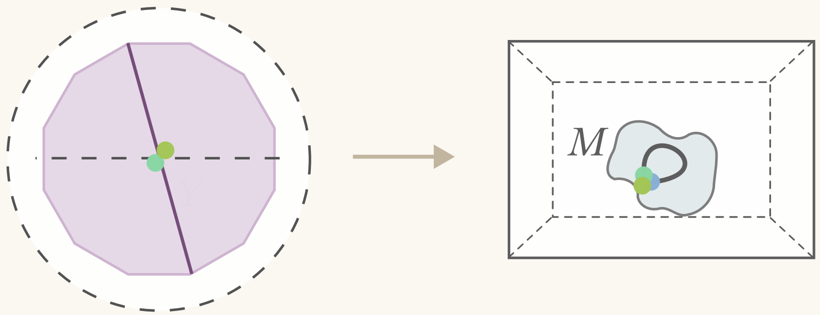

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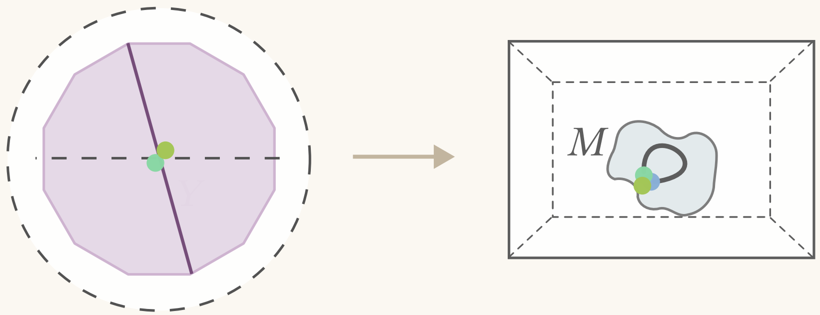

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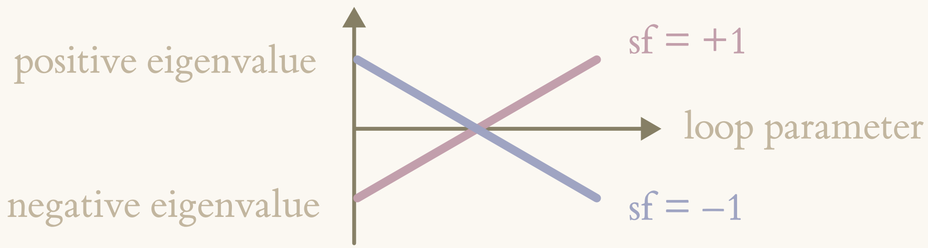

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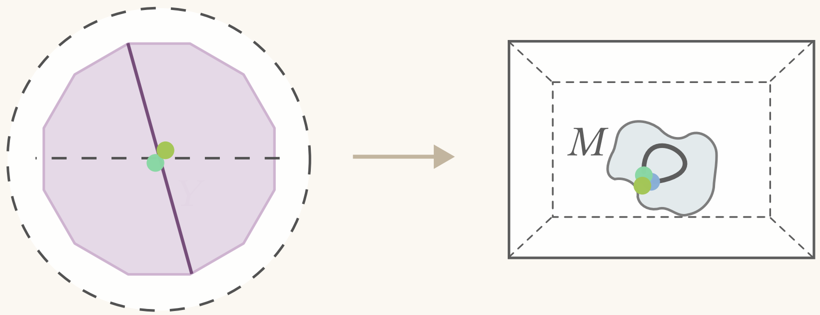

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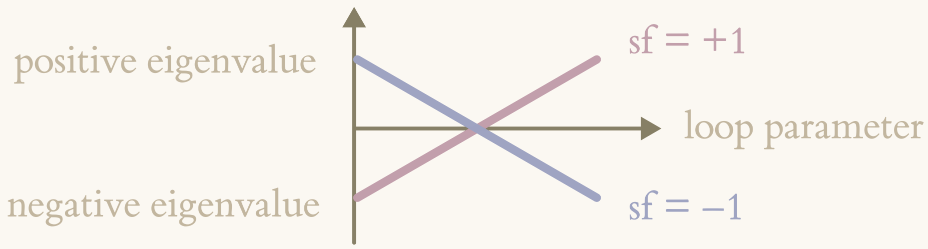

Necessary condition: Around a loop implementing a logical operation, sf vanishes.

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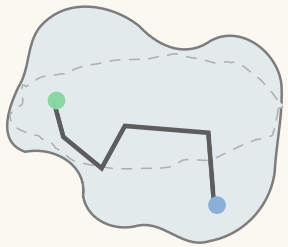

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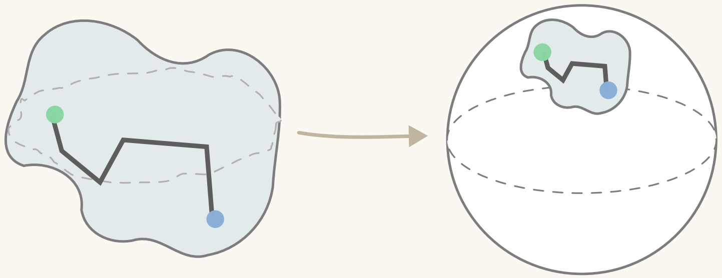

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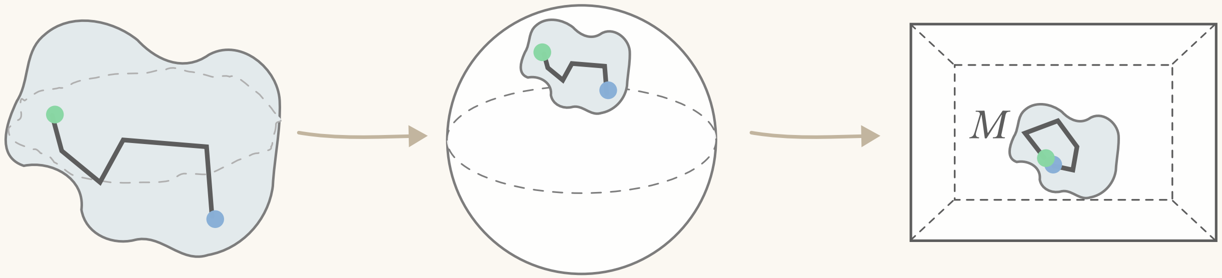

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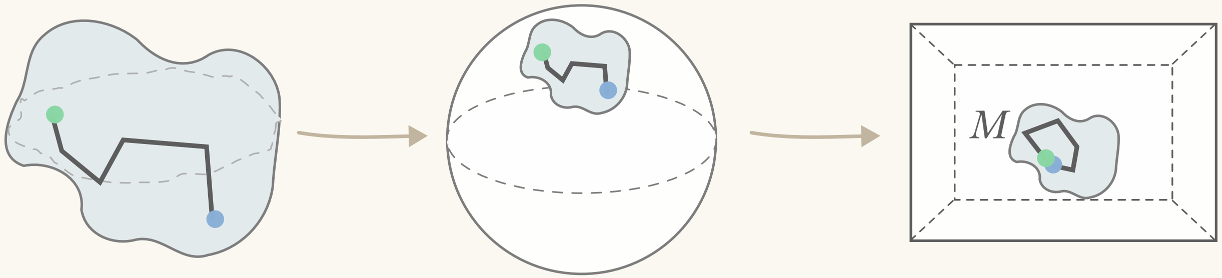

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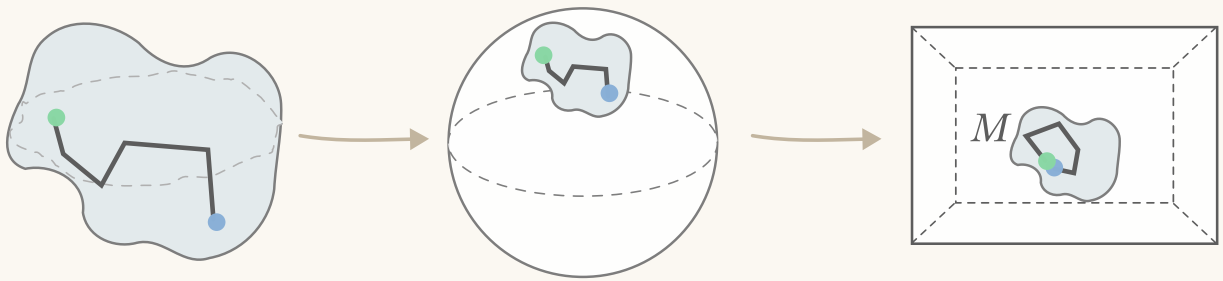

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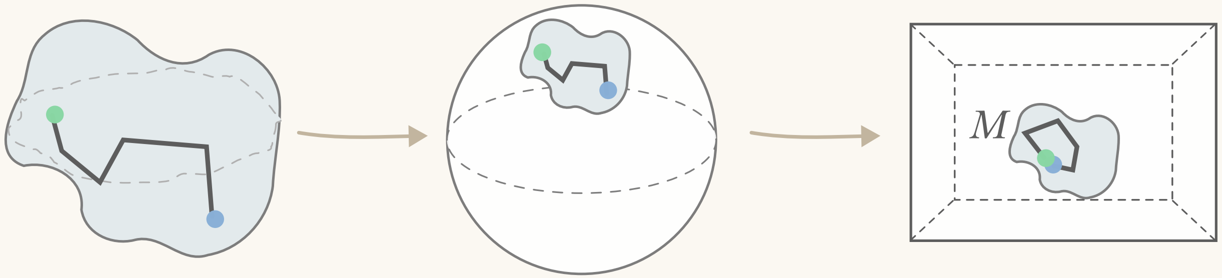

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- It is a marvellous fact that the spectral flow for such a path $\mathrm{sf}(\gamma_i)$ can be exactly determined from a finite-dimensional matrix, namely the *spectral localizer*.

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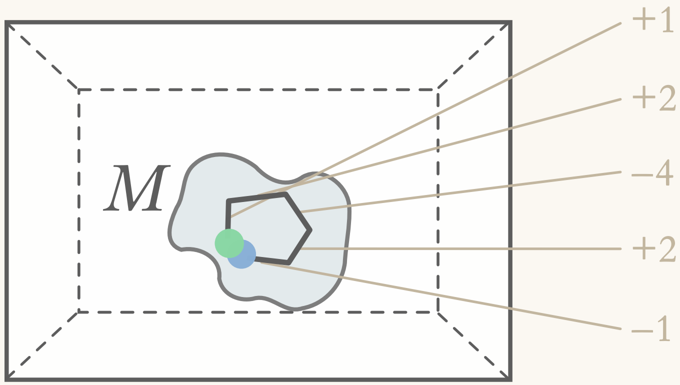

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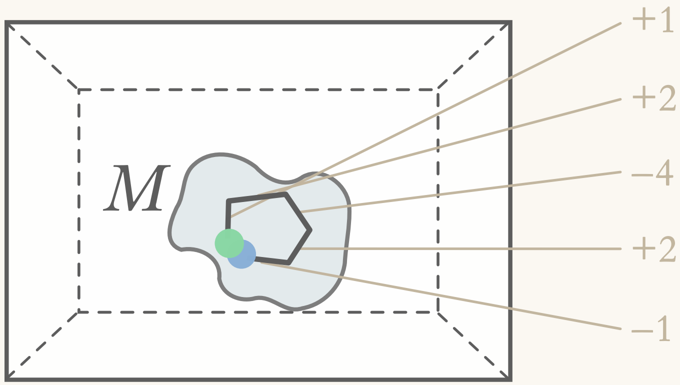


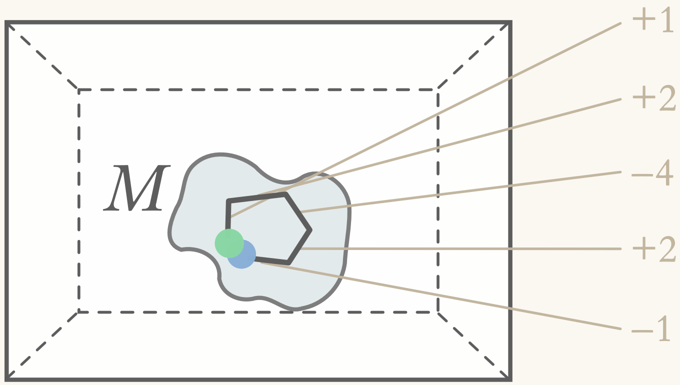
- Thus, we can verify that an operation returns to the code subspace, $\mathrm{sf}(\gamma) = 0$.

Spectral bundles V

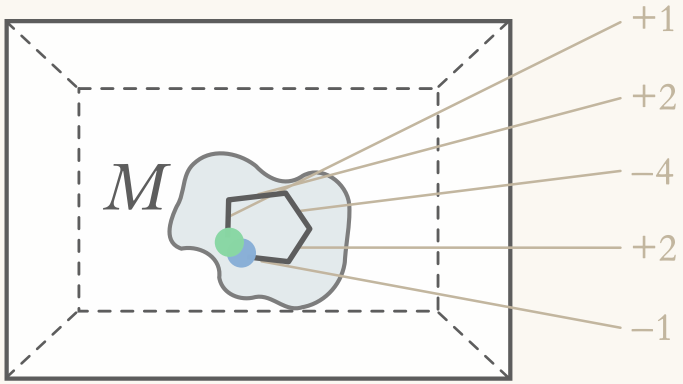
- As shown by Loring, Schulz-Baldes, Cerjan, et al., the spectral flow obeys

$$\mathrm{sf}(\gamma_i) = -\tfrac{1}{2}\,\mathrm{sig}(L_{\boldsymbol{\kappa}}) \quad \text{for} \quad L_\kappa = \begin{bmatrix} \kappa\tilde{D}_i & \tilde{U}_i \\ \tilde{U}_i^\dagger & -\kappa\tilde{D}_i \end{bmatrix},$$

for suitably small κ and approximations on a suitably large Hilbert space $\tilde{H}$.



- Thus, we can verify that an operation returns to the code subspace, $\mathrm{sf}(\gamma) = 0$.
  Cool possibility: certify elements of π_1 (logical operations) from patterns of flow!



Fin